

Mr. Ashok G. Badresiya | A.Y.: 2026-27 | Sem./Year: 1 | Course: 2304CS101 – C Programming - I | Slot Type: Lab (TA) | Division: BCA/BSIT-1

Sr.	Course Content	Planning Title	Planning Description
1	Logic Building Questions (No Code Required)	Lab – 1	Math Logic, Decision Making and Conditions 1. How to check whether a number is odd or even? (A) 2. How will you find the largest of three numbers? (A) 3. How would you find the sum of the digits of any number? (A) 4. How would you check if a number is a palindrome or not? (A) 5. How would you determine if a year is a leap year? (A) 6. How would you find factorial of the given number? (A) 7. How would you check if a number is a prime or not? (A) 8. What logic would you use to print only even numbers between 1 and 100? (B) 9. What logic would you use to print Fibonacci series up to N terms. (C)
2	General Reasoning and Problem Solving	Lab – 2	Daily Logic & Thought Process 1. You have two numbers. Explain how to swap them without using a third variable. (A) 2. How can you count how many times a digit (say 7) appears in a number? (A) 3. How would you check whether the input character is a vowel or consonant? (A) 4. You are given 3 different numbers. How would you sort them in ascending order? (A) 5. You are given a sentence. How do you count the number of words in it? (A) 6. How can you check if a number is already in a list? (B) 7. If a string contains spaces and punctuation, how would you count only the letters? (C)
3	Introduction	Lab – 3	Write an algorithm and Draw Flowchart for following: 1. Calculate addition of two numbers. (A) 2. Calculate average of three numbers. (A) 3. Find area of circle. $\text{pie} \times r \times r$ (A) 4. Find area of triangle. $((\text{height} \times \text{base})/2)$ (A) 5. Calculate simple interest. $(\text{principal} \times \text{roi} \times \text{time period})/100$ (A) 6. Convert temperature from Fahrenheit to Celsius. (Formula: $c = (((f-32) \times 5))/9$) (A)
4	Introduction	Lab – 4	Write an algorithm and Draw Flowchart for following: 1. Convert given feet into inches. $(\text{feet} \times 12)$ (A) 2. Swap two numbers. (Using temporary variable and without using temporary variable) (A) 3. Check whether given number is positive or negative. (A) 4. Check whether the given number is odd or even. (A) 5. Find out largest number from given two numbers. (A) 6. Find out largest number from given three numbers. (B)
5	Introduction	Lab – 5	Write an algorithm and Draw Flowchart for following: 1. Display day name for the given number. (A) 2. Accept three numbers from user and print them in ascending and descending order. (A) 3. Check whether the given year is leap year or not. [If a year can be divisible by 4 but not divisible by 100 then it is leap year but if it is divisible by 400 then it is leap year]. (A)

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			4. Input electricity unit charge and calculate the total electricity bill according to the given condition: - For first 50 units Rs. 0.50/unit - For next 100 units Rs. 0.75/unit - For next 100 units Rs. 1.20/unit - For unit above 250 Rs. 1.50/unit An additional surcharge of 20% is added to the bill. (B)
6	Introduction	Lab – 6	Write an algorithm and Draw Flowchart for following: - Find factorial of given number. (A) - Print all factors of given number. (A) - Print fibonacci series upto given numbers. (A) - Print sum of digit of given number. (A) - Check whether given number is prime or not. (B) - Check whether given number is palindrome or not. (C)
7	Introduction & C Operators & Decision-Making Statements	Lab – 7	Write following programs in C. (Basic) - Print Hello World. (A) - Print your address i) using single printf ii) using multiple printf. (A) - Print addition of 2 numbers. (with & without scanf) (A) - Print average of three numbers. (with & without scanf) (A) - Print area of circle. (pie*r*r) (A)
8	C Operators & Decision-Making Statements	Lab – 8	Write following programs in C. (Type casting and Constant) - Calculate average of 2 integers and show result in float. (A) - Demonstrate const keyword. (A) - ASCII value of character using type casting. (A) - Demonstrate while converting float data into integer data and observe data loss during conversion. (A) - Convert a character into lowercase and uppercase using ASCII code. (A) - Take value from user of a product price and tax in % and print the amount of tax in float. (B)
9	C Operators & Decision-Making Statements	Lab – 9	Write following programs in C. (Basic) - Print area of triangle. ((height*base)/2) (A) - Print simple interest. (principal*roi*time period)/100 (A) - Print temperature from Fahrenheit to Celsius. (Formula: $c = (((f-32)*5))/9$) (A) - Convert seconds into hours, minutes & seconds and print in HH:MM:SS. [e.g. 10000 seconds = 02:46:40] (B) - Convert number of days into year, week & days. [e.g. 375 days mean 1 year, 1 week and 3 days] (B)
10	C Operators & Decision-Making Statements	Lab – 10	Write the following programs in C (Basic) - Calculate perimeter of a Rectangle. (A) - Calculate BMI by taking values of weight and height from user. (Formula: $BMI = weight / (height*height)$) (A) - Calculate electricity bill based on the no. of units consumed and rate. (A) - Calculate total salary after adding a bonus percentage (%). (A) - Calculate total cost of fuel based on distance, milage and fuel price. (Formula: $((distance/milage)*price_per_liter)$) (A)

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			6. Calculate profit % by taking selling price and cost price from user. (B) 7. Calculate final cost of product after taking total cost and discount in % from user. (B)
11	C Operators & Decision-Making Statements	Lab – 11	Write following programs in C. (Decision Making: if, else) 1. Swap two numbers. (Using temporary variable and without using temporary variable) (A) 2. Check whether given number is positive or negative. (A) 3. Check whether the given number is odd or even. (A) 4. Find out largest number from given three numbers. (A) 5. Check whether given character is vowel or consonant. (Using single if only) (B)
12	C Operators & Decision-Making Statements	Lab – 12	Write following programs in C. (Decision Making: Nested and Ladder if) 1. Perform Addition, Subtraction, Multiplication and Division of 2 numbers as per users choice. (A) 2. Enter basic salary of an employee and calculate Gross salary according to given conditions: (A) - Basic Salary ≥ 10000 : HRA = 20% of basic, DA = 80% of basic - Basic Salary ≥ 20000 : HRA = 25% of basic, DA = 90% of basic - Basic Salary ≥ 30000 : HRA = 30% of basic, DA = 95% of basic 3. Check whether the entered character is upper case, lower case, digit or any special character. (A) 4. Input an integer number and check the last digit of number is even or odd. (A) 5. Read marks of five subjects. Calculate percentage and print class accordingly. Fail below 35, Pass Class between 36 to 45, Second Class between 46 to 60, First Class between 61 to 70, Distinction if more than 70. (B)
13	C Operators & Decision-Making Statements	Lab – 13	Write following programs in C. (Decision Making: Nested and Ladder if) 1. Print weekday based on given number. (A) 2. Print month name based on given number. (A) 3. Input electricity unit charge and calculate the total electricity bill according to the given condition: (A) - For first 50 units Rs. 0.50/unit - For next 100 units Rs. 0.75/unit - For next 100 units Rs. 1.20/unit - For unit above 250 Rs. 1.50/unit An additional surcharge of 20% is added to the bill. 4. Determine the roots of the equation $ax^2+bx+c=0$. (B) 5. Three sides of a triangle are entered through the keyboard, WAP to check whether the triangle is isosceles, equilateral, scalene or right-angled triangle. (B) 6. Find the second largest number among three user input numbers. (C)
14	C Operators & Decision-Making Statements	Lab – 14	Write following programs in C. (Decision Making: Switch Case, Conditional Operator) 1. Print day name of week using switch. (A) 2. Print number of days in a month using switch. (A) 3. Perform Addition, Subtraction, Multiplication and Division of 2 numbers as per user's choice using switch. (A) 4. Read 3 numbers, multiply largest number from first two numbers to third one using switch. (B) 5. Check whether character is an alphabet or not using conditional operator. (B) 6. Print number of days in a month considering leap year using switch. (C)

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15	C Operators & Decision-Making Statements	Lab – 15	Write following programs in C. (Decision Making: Switch Case, Conditional Operator) 1. Demonstrate the behaviour of switch case without break. (A) 2. Check whether given number is positive or negative using conditional operator. (A) 3. Find out largest number from given 3 numbers using conditional operator. (A) 4. Check whether number is even number or odd number using conditional operator. (A)
16	C Loops	Lab – 16	Write following programs in C. (While Loop) 1. Print 1 to 10 using While loop. (A) 2. Print 1 to N using While loop. (A) 3. Print odd numbers between 1 to N. (A) 4. Print numbers between two given numbers which is divisible by 2. (A) 5. Print number and its square root for 0 to 9. (B) 6. Calculate the square of integers 1 through 10. (C)
17	C Loops	Lab – 17	Write following programs in C. (While Loop) 1. Print sum of 1 to n numbers. (A) 2. Get 10 numbers from user print count of odd, even numbers. (A) 3. Print all integer greater than 100 and less than 200 that are divisible by 7 but not divisible by 5. (A) 4. Print multiplication table of a given number. (A) 5. Print first 50 numbers in series 1, 4, 7, 10.. (B) 6. Print sum of series $1 + 4 + 9 + 16 + 25 + 36 + n$. (C)
18	C Loops	Lab – 18	Write following programs in C. (While Loop) 1. Find out sum of first and last digit of a given number. (A) 2. Find the sum and average of different numbers which are accepted by user as many as user wants. (A) 3. Find whether the given number is prime or not. (A) 4. Print digits of given number. (A) 5. Check whether the given number is perfect or not. (B) 6. Find whether the given number is prime or not using flag. (B) 7. Check whether the given number is palindrome or not. (C) 8. Check whether the given number is Armstrong or not. (C)
19	C Loops	Lab – 19	Write following programs in C. (Do While Loop) 1. Print sum of 1 to n numbers. (A) 2. Get 10 numbers from user print count of odd, even numbers. (A) 3. Print all integer greater than 100 and less than 200 that are divisible by 7 but not divisible by 5. (A) 4. Print multiplication table of a given number. (A) 5. Print first 50 numbers in series 1, 4, 7, 10.. (B) 6. Print sum of series $1 + 4 + 9 + 16 + 25 + 36 + n$. (C)
20	C Loops	Lab – 20	Write following programs in C. (Do While Loop) 1. Find out sum of first and last digit of a given number. (A) 2. Find the sum and average of different numbers which are accepted by user as many as user wants. (A)

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			3. Find whether the given number is prime or not. (A) 4. Print digits of given number. (A) 5. Check whether the given number is perfect or not. (B) 6. Find whether the given number is prime or not using flag. (B) 7. Check whether the given number is palindrome or not. (C) 8. Check whether the given number is Armstrong or not. (C)
21	C Loops	Lab – 21	Write following programs in C. (For Loop) 1. Print 1 to 10 then modify program print 1 to n. (A) 2. Print sum of 1 to n numbers. (A) 3. Print multiplication table of a given number. (A) 4. Calculate x^y without using power function. (A) 5. Find factorial of the given number. (A) 6. Print all factors of the given number. (B) 7. Print the Fibonacci Series. (B) 8. Count frequency of digits in an integer. (C) 9. Print all ASCII character with their values. (C)
22	C Loops	Lab – 22	Write following programs in C. (Nested For Loop) 1. Print following patterns (A) * ** *** **** ***** 1 12 123 1234 12345 5 54 543 5432 54321 ***** *****

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			*** ** * 12345 1234 123 12 1 2. Print following patterns (B) 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
23	Introduction to array	Lab – 23	Write following programs in C. (Array) 1. Read n numbers from user and print in normal and reverse order. (A) 2. Count number of positive or negative number from an array of n numbers. (A) 3. Count number of even or odd number from an array of n numbers. (A) 4. Count numbers higher than the average of an array. (B) 5. Calculate the average, geometric and harmonic mean of n elements in an array. (C)
24	Introduction to array	Lab – 24	Write following programs in C. (Array) 1. Find Max, Min, Sum, Avg. of given numbers from an array. (A) 2. Read five person height and weight and count the number of person having height greater than 170 and weight less than 50. (A) 3. Sort elements of an array in an ascending order. (A) 4. Count total duplicate elements in an array. (B) 5. Find missing numbers of sequence using array.(in a sequence 1,2,4,5,7,8,10, Missing numbers are 3,6,9) (C)
25	Introduction to array	Lab – 25	Write following programs in C. (Array) 1. Copy all elements of one array to another. (A)

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			2. Count total number of negative elements in array. (A) 3. Count number of elements divisible by 3 in array. (A) 4. Delete all duplicate elements from an array. (B) 5. Reverse elements of an array without using second array. (B) 6. Find two largest elements in a one dimensional array. (C)
26	Introduction to array	Lab – 26	Write following programs in C. (Array) 1. Search element in array. (A) 2. Find maximum and minimum value from array. (A) 3. Input a string in character array and print string and length of string. (A) 4. Swap first element with last, second to second last and so on. (B) 5. Insert new value in the sorted array. (C)
27	2-D array and String	Lab – 27	Write following programs in C. (2D Array) 1. Read values in two-dimensional array and print them in matrix form. (A) 2. Count number of positive, negative and zero elements from 3 X 3 matrix. (A) 3. Read and store the roll no and marks of 20 students using 2D array. (A) 4. Print Transpose of a matrix. (B) 5. Perform Multiplication of two matrices. (B)
28	2-D array and String	Lab – 28	Write following programs in C. (2D Array) 1. Count number of odd and even elements from N X N matrix. (A) 2. Check given two matrices are identical or not. (A) 3. Read a matrix and print diagonal elements and its sum. (B) 4. Check a given matrix is a sparse matrix or not. (B) 5. Print the upper triangular matrix. (C)
29	2-D array and String	Lab – 29	Write following programs in C. (String) 1. Use string handling functions strlen(), strcmp(), strcpy(), strcat(), strrev(), strlen() andstrupr() (A) 2. Print all character of given string. (A) 3. Find a character from given string. (A) 4. Replace a character in given string. (A)
30	2-D array and String	Lab – 30	Write following programs in C. (String) 1. Find length of given string without using built-in function. (A) 2. Convert a string into upper case and lower case string without using built-in function. (A) 3. Copy given string into another string without using built-in function. (A) 4. Read two string from user and merge second string into first string without using built-in function. (B) 5. Count occurrence of a given character in a string. (B)